Data Analytics Insights Report

This report explains the main findings from the four tasks. We have used SQL queries, Excel and Power BI to study the traffic patterns. The main aim of this project is to help the smart-city mobility team understand the busy periods and make better planning decisions.

The dataset contains 48204 records and 40575 different recorded hours. Some hours appear more than once. I have used all records, while the total hours card counts each recorded hour once.

Task 1 Yearly traffic and holidays

Recorded traffic totals fell in 2014 and 2015, then increased in 2016 and 2017. The largest decrease was 44.17% between 2013 and 2014. The highest recorded total was in 2017, at about 35.43 million. This was 20.12% higher than 2016.

However, all the years do not have the same amount of data. For example, 2012 starts in October, and there are missing periods in other years. Repeated hours also affect the totals. The team should check these gaps before using the results to say that actual traffic increased or decreased.

The holiday results did not show a clear link between temperature and traffic. The record labelled New Year's Day was 4.68C warmer in 2017 than in 2016, but traffic fell from 1513 to 798. Labor Day also showed no consistent pattern. These results only cover midnight records, so they cannot describe the whole holiday. No New Year's Day record was availble for 2015.

Task 2 Traffic statistics and temperature

The average traffic volume was 3259.82, and the median was 3380. The population standard deviation was 1986.84, and the population variance was 3947533.43. Traffic values ranged from 0 to 7280, giving us a range of 7280.

These results show that traffic changes a lot between records. The average alone does not show how busy or quiet a particular hour can b e. The team should study these periods seperately.

The correlation between temperature and traffic was 0.1303. This shows a weak positive link. Warmer temperatures sometimes came with more traffic, but the relationship was small. This does not prove that temperature causes traffic to increase. Time of day and people's travel habits might also explain the pattern.

Task 3 Probability and congestion

Congestion was defined as traffic volume above 5500. Using this rule:

- 14.73% of records had congestion.

- 27.78% had clear weather.
- 3.66% had both congestion and clear weather.
- Among congestion records, 24.83% had clear wether and 26.30% had temperatures above 292K.

The independence calculations did not give equal results. This indicates that the events were not exactly independent in this dataset.

The odds ratio for congestion in clear weather compared with cloudy weather was 0.7354. Congestion had lower odds in clear weather. However, this does not prove that weather caused the difference.

Traffic volume aso does not tell us how slowly vehicles were moving.

Task 4 Power BI dashboard

The original data had 9 columns and no blank or null values. We checked the data types and added Hour, Temperature in Celsius, Traffic Category and Datecolumns.

The daily chart shows frequent changes in traffic. In 2017, the highest hourly average was at 4 PM, at about 5798. The lowest was at 3 am, around 382.

Clouds had the highest average traffic at 3618.45. Squall had the lowest at 2061.75. The difference was 1556.70.

The scatter plot showed no clear temperature–traffic pattern. We havce found 10 invalid(outliers)temperature readings at −273.15°C and hid them in the second plot.

The cards showed 40575 recorded hours, average traffic of 3259.82 and average temperature of 8.06C. The temperature average still includes the invalid readings. All three filters worked correctly in our combined test.

Summary

The team should focus monitoring on busy afternoon hours and consider quieter hours for road maintenance. It should also check missing hours, , invalid temperatures , repeated records and zero traffic values before further analysis. Weather might help with planning, but it should be considered along with time of day and other travel information.